Part 0

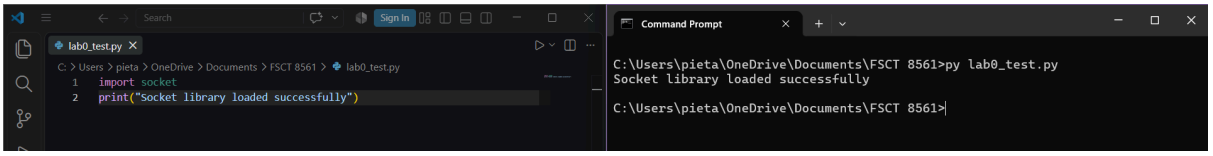
```
C:\Windows\System32>py --version
Python 3.11.5

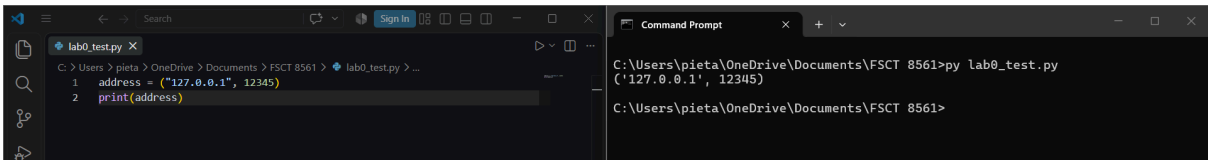
C:\Windows\System32>
```

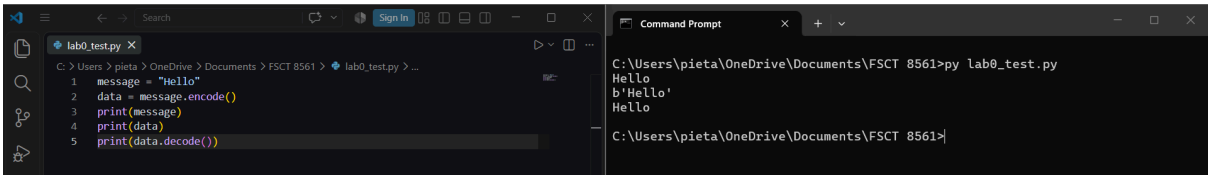
```
C:\Windows\System32>py
Python 3.11.5 (tags/v3.11.5:cce6ba9, Aug 24 2023, 14:38:34) [MSC v.1936 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print ("Hello FSCT 8561")
Hello FSCT 8561
>>> exit()

C:\Windows\System32>
```

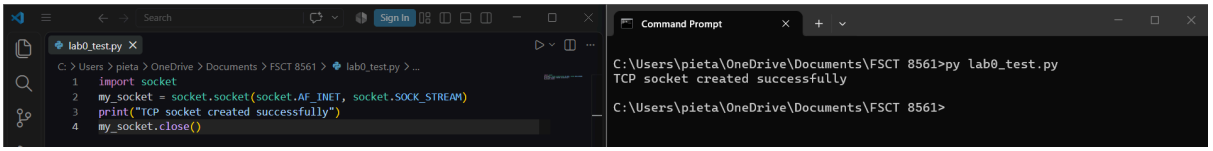
Part 1




Part 2

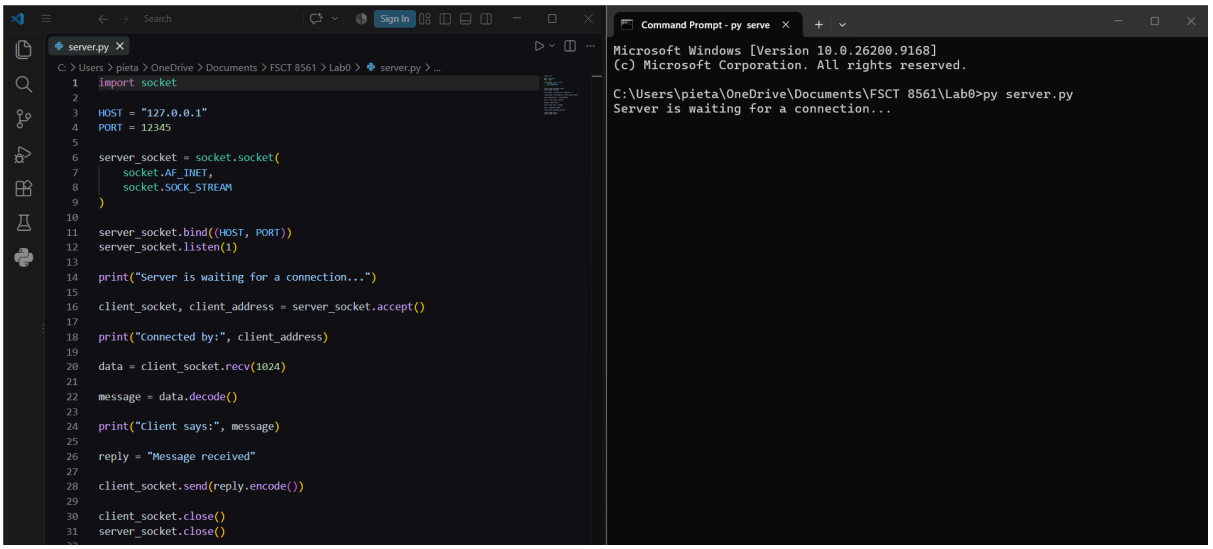



**Reflection Question 1**

Explain in your own words:

1. What is a socket?
    - A **socket** is an endpoint for sending or receiving data across a computer network.
2. What does AF_INET mean?
    - AF_INET stands for Address Family: Internet
3. What does SOCK_STREAM mean?
    - SOCK_STREAM means the socket will use a stream-based, connection-oriented protocol

## Part 4



```
import socket

HOST = "127.0.0.1"
PORT = 12345

server_socket = socket.socket(
    socket.AF_INET,
    socket.SOCK_STREAM
)

server_socket.bind((HOST, PORT))
server_socket.listen(1)

print("Server is waiting for a connection...")

client_socket, client_address = server_socket.accept()

print("Connected by:", client_address)

data = client_socket.recv(1024)

message = data.decode()

print("Client says:", message)

reply = "Message received"

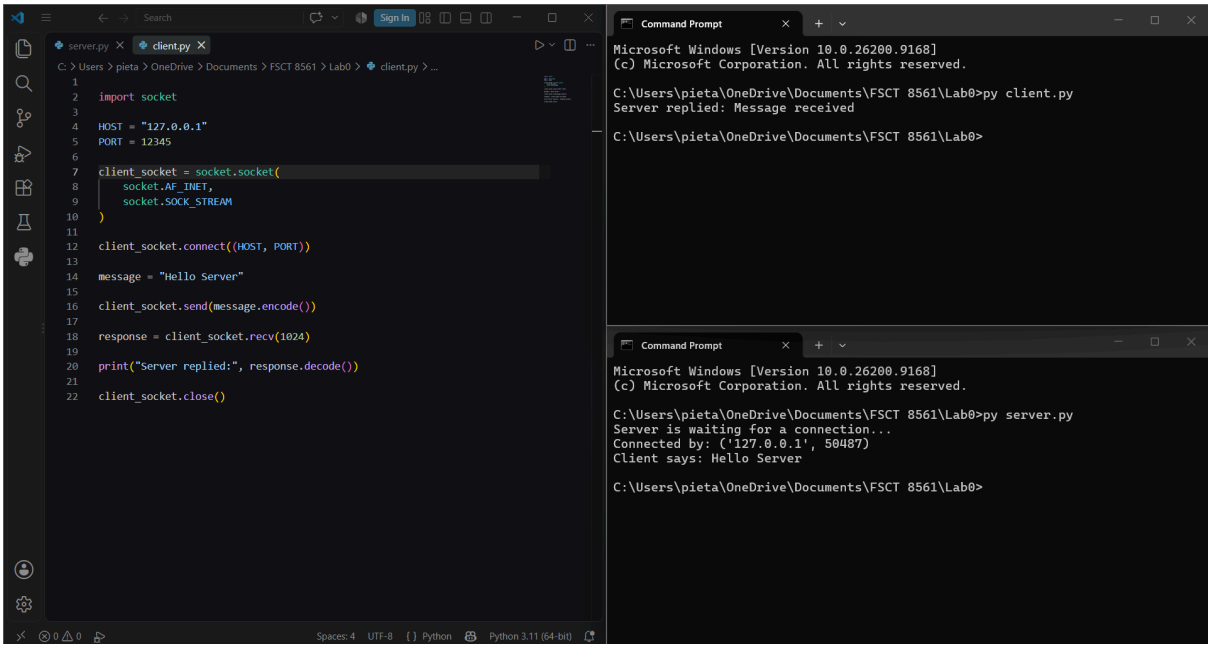
client_socket.send(reply.encode())

client_socket.close()
server_socket.close()
```

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>py server.py
Server is waiting for a connection...
```

## Part 5



```
import socket

HOST = "127.0.0.1"
PORT = 12345

client_socket = socket.socket(
    socket.AF_INET,
    socket.SOCK_STREAM
)

client_socket.connect((HOST, PORT))

message = "Hello Server"

client_socket.send(message.encode())

response = client_socket.recv(1024)

print("Server replied:", response.decode())

client_socket.close()
```

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>py client.py
Server replied: Message received

C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>
```

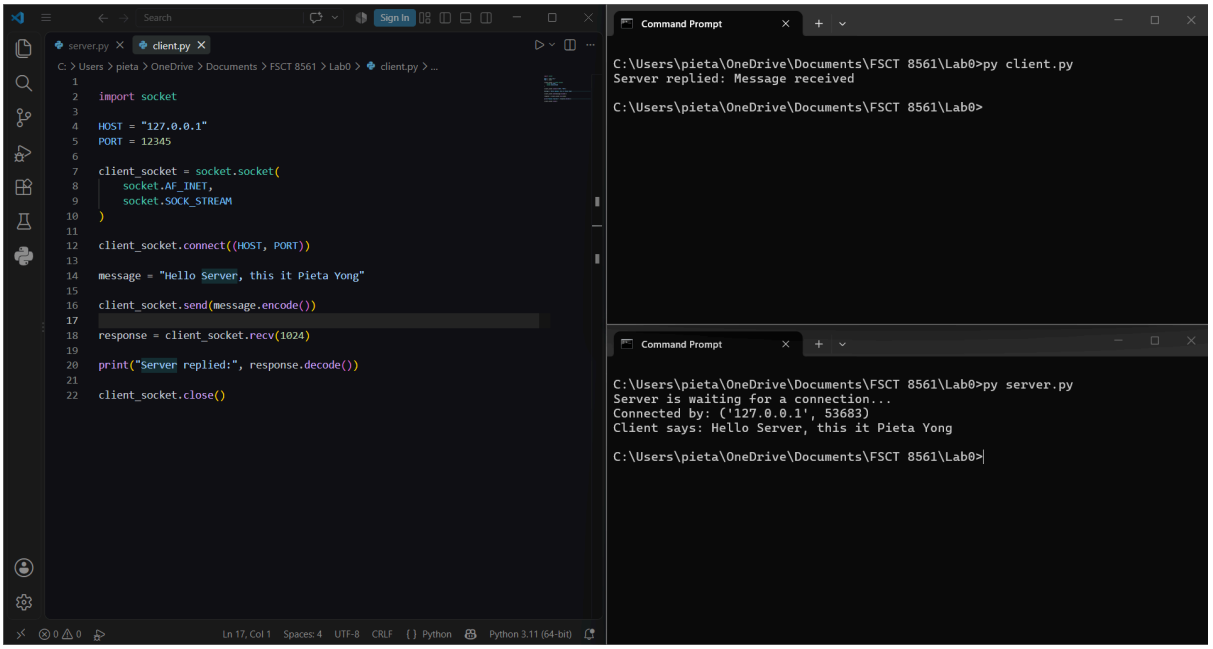
```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>py server.py
Server is waiting for a connection...
Connected by: ('127.0.0.1', 50487)
Client says: Hello Server

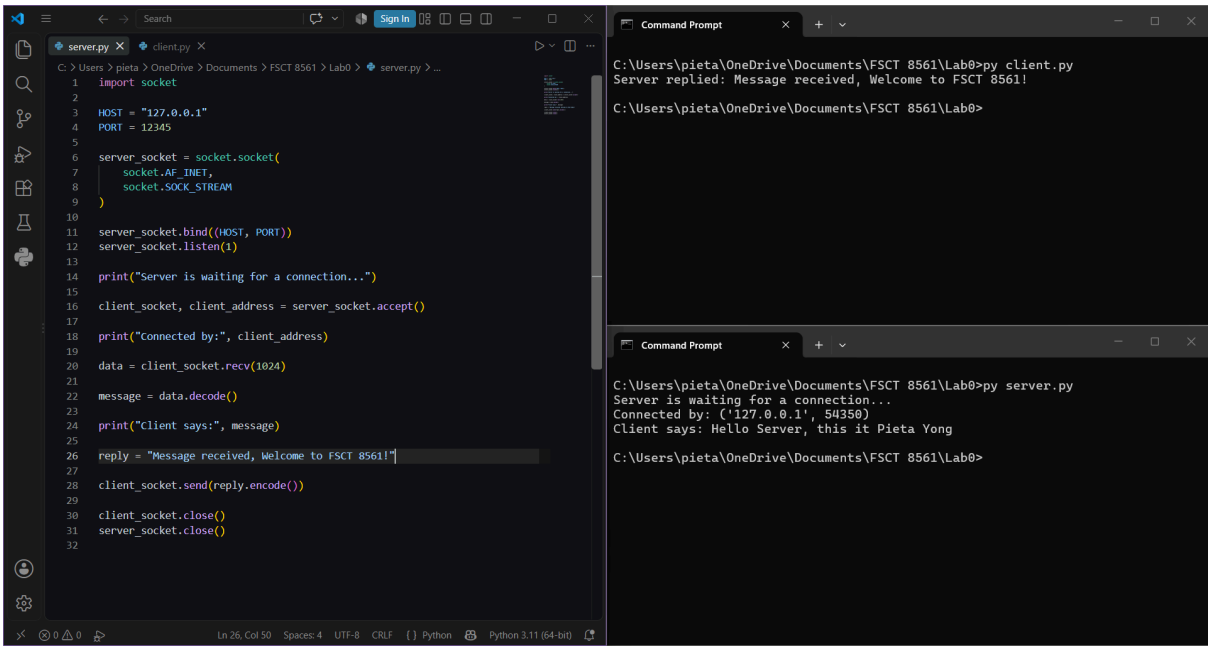
C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>
```

Part 6

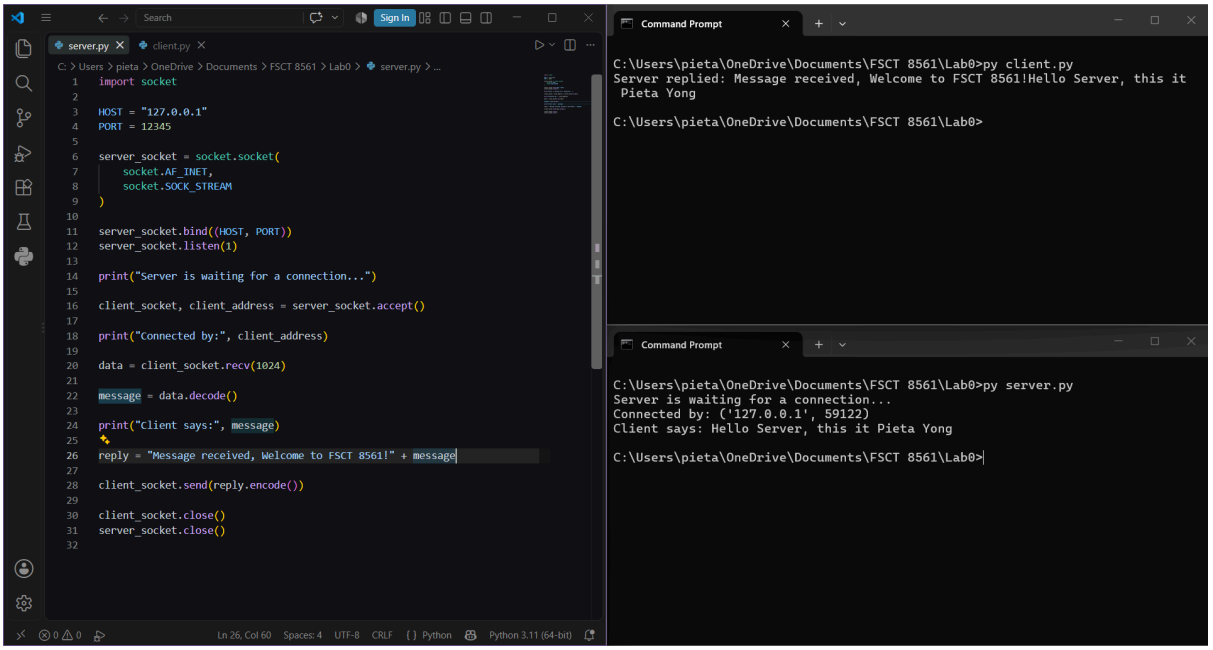
Task A - “Hello this is Pieta Yong”




Task B - "Message received, Welcome to FSCT 8561!"



Task C "Server received: " + message



```
# server.py
import socket

HOST = "127.0.0.1"
PORT = 12345

server_socket = socket.socket(
    socket.AF_INET,
    socket.SOCK_STREAM
)

server_socket.bind((HOST, PORT))
server_socket.listen(1)

print("Server is waiting for a connection...")

client_socket, client_address = server_socket.accept()

print("Connected by:", client_address)

data = client_socket.recv(1024)

message = data.decode()

print("Client says:", message)

reply = "Message received, Welcome to FSCT 8561!" + message

client_socket.send(reply.encode())

client_socket.close()
server_socket.close()
```

```
Command Prompt

C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>py client.py
Server replied: Message received, Welcome to FSCT 8561!Hello Server, this it
 Pieta Yong

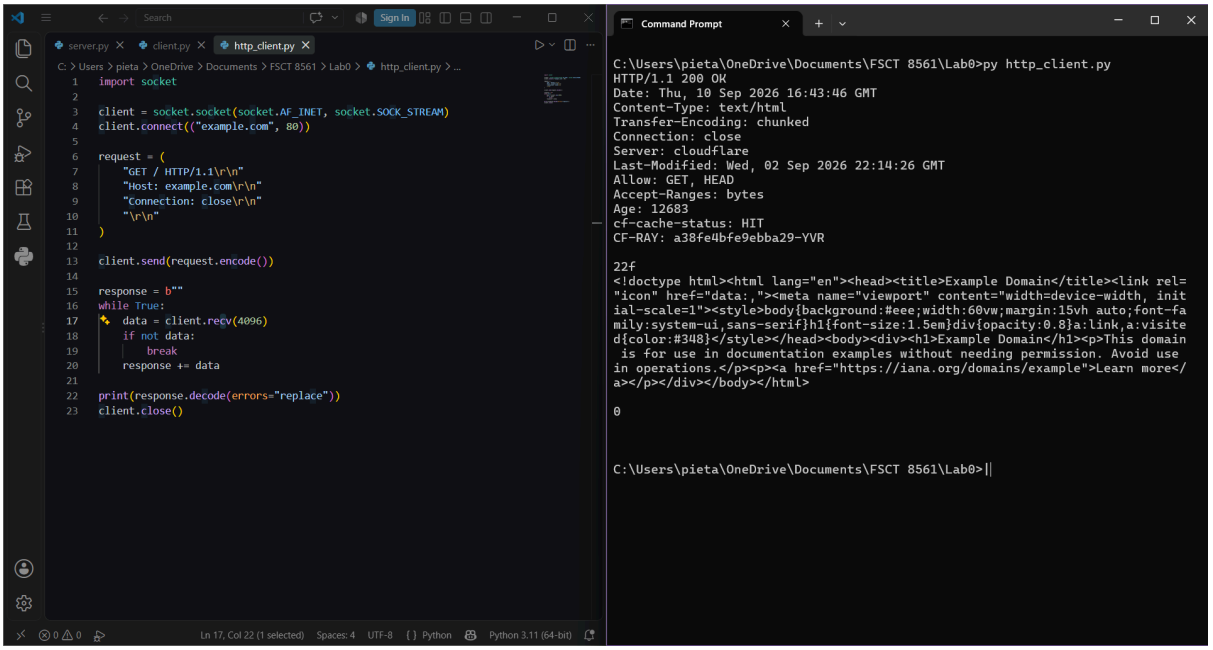
C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>
```

```
Command Prompt

C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>py server.py
Server is waiting for a connection...
Connected by: ('127.0.0.1', 59122)
Client says: Hello Server, this it Pieta Yong

C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>
```

Part 7



```
# http_client.py
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("example.com", 80))

request = (
    "GET / HTTP/1.1\r\n"
    "Host: example.com\r\n"
    "Connection: close\r\n"
    "\r\n"
)

client.send(request.encode())

response = b""
while True:
    data = client.recv(4096)
    if not data:
        break
    response += data

print(response.decode(errors="replace"))
client.close()
```

```
Command Prompt

C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>py http_client.py
HTTP/1.1 200 OK
Date: Thu, 10 Sep 2026 16:43:46 GMT
Content-Type: text/html
Transfer-Encoding: chunked
Connection: close
Server: cloudflare
Last-Modified: Wed, 02 Sep 2026 22:14:26 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 12683
cf-cache-status: HIT
CF-RAY: a38fe4bfe9ebba29-YVR

22f
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel=
"icon" href="data:,"><meta name="viewport" content="width=device-width, init
ial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-fa
mily:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visite
d{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain
 is for use in documentation examples without needing permission. Avoid use
in operations.</p><p><a href="https://iana.org/domains/example">Learn more</
a></p></div></body></html>

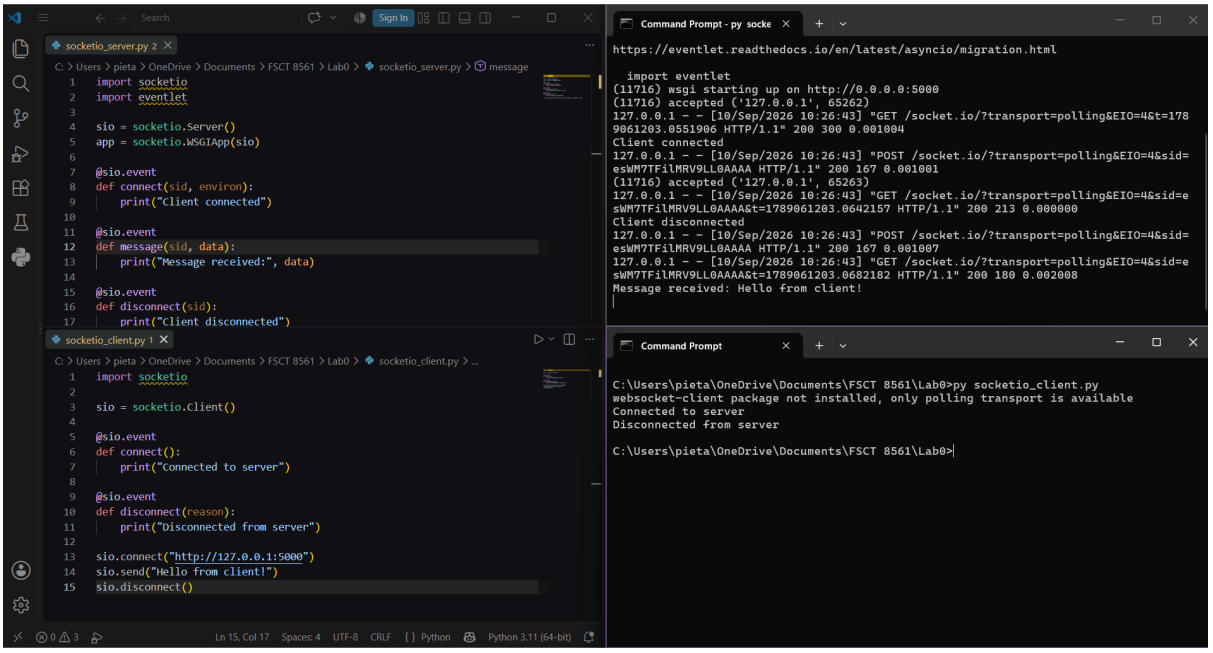
0


C:\Users\pieta\OneDrive\Documents\FSCT 8561\Lab0>
```

**Reflection Question 2**

1. Which host did the client connect to?
    - To the host at  Example Domain
2. Which port was used?
    - 80
3. Why is port 80 relevant?
    - Port 80 is used for standard HTTP web traffic
4. What did the server send back?
    - The server sent an HTTP/1.1 200 OK response status along with headers

Part 8




**Security Reflection**

1. What could happen if a server accepts connections from anyone?
    - Unauthorized users could connect
2. Is the Lab 0 message encrypted?
    - No
3. Could network traffic be observed by another system?
    - Yes because it is not encrypted
4. Why should servers validate input?
    - to prevent invalid or malicious data from causing errors or security issues
5. Why might authentication be useful?
    - helps verify that only authorized users or clients can connect to the server
6. Suggest two ways to make this application more secure.
    - encrypt the connection using TLS/HTTPS
    - require authentication before allowing clients to communicate with the server.

Part 9

4. A short paragraph explaining what you learned and what was unclear

- From this lab I learned how to create a simple TCP server and client and demonstrate basic TCP connections to send and receive messages. I think that as someone with “little or no previous Python experience”, It would have been helpful to know what chapters/sections from the readings were required to complete this lab.

5. A link to your private GitHub repository containing the source code for Lab 0.
https://github.com/pietayng/FSCT8561-Labs-Pieta-Yong/invitations